SPRING 2020

Economic Mathematics II
FINAL EXAM

Notes: (1) this is an open-book/notes exam; (2) there are 9 questions, with a total of 100 points; (3) suggestion: have a look at all problems and first solve the problems you feel easiest; (4) good luck!

1. **[12 pts]** Multiple Choice (**Each question has ONE correct answer**)

(a) [3 pts] If random variables (ξ, η) are both normally distributed, $\mathrm{E}(\xi\eta) = \mathrm{E}(\xi)\mathrm{E}(\eta)$ is which of following condition for the independence of ξ and η?
(A) Sufficient condition
(B) Necessary condition
(C) Neither sufficient or necessary condition
(D) Sufficient and necessary condition

(b) [3 pts] Roll a die n times repeatedly, and let ξ and η represent the times of odd and even numbers respectively, then the correlation coefficient of ξ and η is equal to
(A) -1
(B) 0
(C) 0.5
(D) 1

(c) [3 pts] Suppose $X_1, \cdots, X_n, X_{n+1}, \cdots, X_{n+m}$ is a sample independently drawn from the population with normal distribution $N(0, \sigma^2)$, then the distribution of the statistic

$$U = \frac{m \sum_{i=1}^{n} X_i^2}{n \sum_{i=n+1}^{n+m} X_i^2}$$

follows
(A) $F(m, n)$
(B) $F(n-1, m-1)$

(C) $F(n, m)$
(D) $F(m-1, n-1)$

(d) [3 pts] Suppose $\{X_1, \cdots, X_n\}$ is a i.i.d. sample that follows $N(\mu, \sigma^2)$, then the statistic $\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2$ follows
(A) $\chi^2(n)$
(B) $\chi^2(n-1)$
(C) $N(\mu, \sigma^2/n)$
(D) $t(n-1)$

2. **[8 pts]** Suppose X has a probability density function (PDF)

$$f_X(x) = \begin{cases} |x|, & \text{if} -1 < x < 1 \\ 0 & \textit{otherwise.} \end{cases}$$

Let $Y = X^2$.

Find $\text{cov}(X, Y)$. Give your reasoning.

3. **[8 pts]** Suppose $E(Y|X) = 2$. Find $\text{cov}(X, Y)$. Give your reasoning.

4. **[8 pts]** Suppose X and Y are random variables such that $\text{var}(X) = 9, \text{var}(Y) = 4$ and the correlation $\rho_{XY} = -1/6$. Determine $\text{var}(X - 3Y + 4)$. Give your reasoning.

5. **[14 pts]** Chicago's minimum wage rose from \$4.25 to \$5.05 per hour on April 1, 2000. As an economist, you are hired by the state government to evaluate the impact of the law on employment. Suppose you surveyed 320 fast-food restaurants in Chicago before and after the law. The average change in employment is 0.59 and the standard deviation is 9.67. Suppose you want to test whether full-time employees per restaurants changed before and after the law in Chicago.

   (a) [5 pts] Conduct a two-sided test at the 1% significance level. Do you think the law on minimum wage affect the full-time employees in Chicago? Draw a conclusion by explicitly writing down hypotheses and test statistic.

   (b) [2 pts] What is the interpretation of the 1% significance level in Part (a)?

   (c) [5 pts] What is the 95% confidence interval estimate for the change in full-time employees in Chicago? Give your reasoning.

(d) [2 pts] What is the interpretation of the 95% confidence interval estimate in Part (c)?

6. **[15 pts]** Suppose $\mathbf{X}^n = (X_1, X_2, ..., X_n)$ is an IID $N(0, \sigma^2)$ random sample. Define two estimators

$$S_n^2 = (n-1)^{-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2,$$

and

$$\hat{\sigma}_n^2 = n^{-1} \sum_{i=1}^{n} X_i^2.$$

where $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$.

(a) [7 pts] Check whether S_n^2 and $\hat{\sigma}_n^2$ are unbiased estimators for σ^2 respectively. Give your reasoning.

(b) [8 pts] Which estimator is more efficient? Give your reasoning.

7. **[15 pts]** Suppose there are k independent random samples from a normal $N(\mu, \sigma^2)$ population, but each sample has a different sample size. Let S_i^2 be the sample variance of the i-th random sample whose sample size is n_i.

(a) [10 pts] Define a class of estimators for σ^2 as follows: $\bar{S}^2 = \Sigma_{i=1}^{k} c_i S_i^2$. Find the best unbiased estimator of σ^2 from this class. Denote this best unbiased estimator as $\bar{S}^{*2}$. Give your reasoning.

(b) [5 pts] Define a pooled sample variance S^2 by pooling k random samples together. That is, S^2 is the sample variance from the pooled random sample that combining all k independent random samples (so that the total sample size $n = \Sigma_{i=1}^{k} n_i$.) Which estimator, $\bar{S}^{*2}$ or S^2, is better? Give your reasoning.

8. **[10 pts]** Suppose $\{X_i\}_{i=1}^{n}$ is an i.i.d. random sample with the following pdf

$$f_X(x) = \theta x^{-\theta-1},$$

for $x \geq 1$ and 0 otherwise. $\theta > 0$ is an unknown parameter. Please show that

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} \ln X_i$$

is a unbiased estimator for $1/\theta$.

9. **[10 pts]** Suppose $\{X_i\}_{i=1}^n \sim i.i.d.N(\mu, \sigma^2)$. Let $T = T(X_1, X_2, \cdots, X_n)$ is a linear unbiased estimator for μ. Please show that the correlation coefficient between the sample mean $\bar{X}_n$ and T is $\frac{\sigma}{\sqrt{nVar(T)}}$, where $Var(T)$ is the variance of T.